

MUHAMMAD AHMAD

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PROFESSIONAL SUMMARY

AI/ML Engineer and Data Science student building end-to-end ML, NLP, and LLM systems from raw data to deployed, explainable models. Hands-on experience with RAG pipelines, model deployment, data warehousing, and BI analytics using Python, PyTorch, HuggingFace, and LangChain.

EDUCATION

FAST-NUCES, Lahore <i>Bachelor of Science in Data Science</i> Punjab College, Lahore <i>FSc Pre-Engineering</i>	2023 – Present Dean's List — Spring 2025 2021 – 2023
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EXPERIENCE

Nepta Solutions <i>Technical Intern</i> - Developed backend automation workflows using C# and SAGE 50 for business process integration. - Integrated RESTful API endpoints for system configuration and data exchange; collaborated via Microsoft Teams.	Jun 2025 – Aug 2025 United Kingdom (Remote)
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PROJECTS

PakWheels Used Car Price Predictor | *Python, XGBoost, LightGBM, SHAP, Streamlit*
– Scraped 24,000+ listings, built a 7-step cleaning pipeline (97.6% retention), and engineered 12 domain features.
– Trained XGBoost/LightGBM with SHAP-based explainability; deployed end-to-end via Streamlit.

Retail Store Inventory Forecasting | *PostgreSQL, Power BI, ETL, DAX*
– Designed a Star Schema data warehouse for 73,000+ records with ETL pipelines achieving 98% query speedup.
– Built Power BI dashboard with DAX to analyze demand, pricing elasticity, and seasonal trends.

Emotion Detection in Social Media Text | *NLP, Flan-T5, DistilBERT, HuggingFace*
– Compared zero-shot, few-shot, and fine-tuning on 20,000 Reddit comments; fine-tuned DistilBERT for multi-class emotion classification.

AI-Powered Support Ticket Triage System | *Python, Flask, ChromaDB, Llama 3.3, Groq API*
– Built an end-to-end automation pipeline utilizing a local **ChromaDB**-backed RAG architecture and structured Prompt Engineering to dynamically classify incoming support tickets.
– Designed a system to auto-generate grounded, policy-compliant draft responses, minimizing manual support triage time.

Electronics Inventory Management Dashboard | *SQL, Power BI*
– Built a Power BI dashboard backed by SQL covering sales performance, stock levels, supplier analysis, and regional sales tracking.

Sales Trend Analysis | *Python, Data Mining, Time-Series Analysis*
– Performed time-series analysis on sales data to identify seasonal patterns and demand trends using data mining techniques.

SKILLS

Programming: Python, SQL, C, C++, JavaScript
ML / AI: Regression, Classification, Clustering, Dimensionality Reduction, Web Scraping
NLP / LLMs: HuggingFace, Transformers, Flan-T5, DistilBERT, Fine-tuning, Zero-shot, Few-shot, RAG, LangChain.js, Pinecone, Vector Databases, Groq Cloud API, Prompt Engineering
Deep Learning: PyTorch (ANN, RNN, CNN), TensorFlow, Keras
Computer Vision: OpenCV, Image Processing, CNN-based Models
Explainable AI: SHAP, XAI, Feature Importance
Data Mining: Association Rules, Pattern Discovery, Clustering, Time-Series Analysis
API & Backend Integration: API Integration, Node.js, Express
Databases & Backend: PostgreSQL, SQL Server, MongoDB, Data Warehousing, ETL, SQL Optimization
EDA / Analysis: Exploratory Data Analysis, Data Cleaning, Statistical Analysis, Scikit-learn
Visualization & UI: React, Power BI (DAX), Tableau, Matplotlib, Seaborn
AI Tools: ChatGPT, Claude, GitHub Copilot, Cursor, Antigravity
Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Streamlit, Vercel, GitBash

CERTIFICATIONS

Feature Engineering for ML in Python — *DataCamp*
IBM Machine Learning — *Coursera*
Introduction to Deep Learning with PyTorch — *DataCamp*
Relational Database Design — *Coursera*